JUAN SEBASTIÁN SÁNCHEZ URBANO

Industrial Automation Engineer · Machine Learning, Technical Sales, Power BI and AI

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LinkedIn · GitHub · Portfolio

PROFESSIONAL PROFILE

Industrial Automation Engineer focused on Machine Learning, technical sales and data analysis. Since October 2026 I have been a technical advisor in the fastener sector, where I advise customers on product selection and technical solutions. Previously I worked as a technical and sales advisor at a technical-solutions company, where I analyzed customer information and automated data processes, and as CNC and Robotics area leader I analyzed production and quality-control variables. I build predictive models, Power BI dashboards and AI workflows with Python, R and n8n, and design medallion-architecture (Bronze/Silver/Gold) ETL pipelines. I combine industrial engineering with analytics to turn data into sales and production decisions.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning, Predictive modeling, Classification, Statistical analysis, EDA, AI agents, AI chatbots, Python, R.

Sales & Business Intelligence: Technical and commercial advisory, Sales analysis, Power BI, Advanced Excel, Dashboards, Data consolidation.

Data engineering & Cloud: ETL pipelines, Medallion architecture, Advanced SQL, MySQL, AWS, REST APIs, Data quality and cleaning.

Automation & Industrial engineering: n8n, Industrial automation, PLC programming, CNC and robotics, Real-time IoT sensors.

Web development & deployment: HTML, CSS, JavaScript, Bash, YAML, Git and GitHub, GitHub Actions, GitHub Pages.

Methodology & Management: PDCA (continuous improvement), Analytical problem solving, Results orientation, Effective communication.

PROFESSIONAL EXPERIENCE

Technical Advisor — Industrias CATO

Oct 2026 – Present

- I advise customers on fastener selection: threads, heads, materials, finishes and applications, and support the technical validation of their requirements.
- **Achievement:** I built technical reference tools available in this portfolio: a fastener academy and a thread calculator.

Technical, Sales and Data Automation Advisor — Proeléctricos

2019 – Nov 2025 · Mar 2026 – Sep 2026

- I analyze and validate customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** I designed and implemented an AI chatbot for the web channel, integrated into an n8n automation pipeline that automatically extracts, validates and routes reports and information to the responsible areas.

CNC and Robotics Area Lead (Production Data Analysis) — Metálicas Mundial

Oct 2025 – Jan 2026

- I led the capture and analysis of critical production and quality-control variables, ensuring the reliability of manufacturing process data.
- **Achievement:** I identified bottlenecks and optimized cycle times by analyzing machinery operating data under agile continuous-improvement methodologies, increasing the area's productivity.

FEATURED PROJECTS

Immersive environments for phobia identification

Bachelor's thesis · September 2025 · Unity, Meta Quest 2, Realidad virtual

- I designed and programmed immersive environments in Unity for Meta Quest 2, focused on identifying phobias such as fear of fire and heights.
- **Impact:** I proposed a virtual-reality system that analyzes user reactions to controlled stimuli, for fear assessment and mental-health applications.

Mobile robot control through virtual reality

CINVESTAV, Mexico City · February 2026 · Unity, C#, ROS, Meta Quest 2

- Development of a mobile-robot control system through virtual reality, using Unity and Meta Quest 2, in the Automatic Control Area of CINVESTAV.
- **Impact:** Integration of serial communication, IMU and LIDAR sensors, 3D modeling and robot control from the virtual-reality environment.

EDUCATION

Industrial Automation Engineering

Universidad del Cauca · Graduated 2025

Complementary training

- Artificial Intelligence Bootcamp (MinTIC / Talento Tech) · Data Science, Python and Machine Learning (Platzi)